Acid and Bases Past Paper Practice

Wednesday, September 30, 2026 6:36 PM

1. During a titration a learner adds $25cm^3$ of $NaOH(aq)$ of concentration $0.1\ mol.dm^{-3}$ to an Erlenmeyer flask and titrates this solution with $H_2SO_4(aq)$ of concentration $0.1\ mol.dm^{-3}$. The balanced equation for the reaction that takes place:

$$2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

1.1 Determine the volume of $H_2SO_4(aq)$ which must be added to neutralise the $NaOH(aq)$ in the Erlenmeyer flask.

$V(NaOH) = 0.025\ dm^3$

$NaOH: c = \frac{n}{V} \Rightarrow n = c \times V = 0.1 \times 0.025 = 0.0025\ mol$ *

$2NaOH : H_2SO_4$

$2 : 1$ (1.25X10⁻³)

$0.0025 : 0.00125\ (mol)$

$H_2SO_4: c = \frac{n}{V} \Rightarrow \frac{c \times V}{c} = \frac{n}{c} \Rightarrow V = \frac{n}{c}$

$V = \frac{n}{c} = \frac{0.00125}{0.1} = 0.0125\ dm^3$

If the learner passes the endpoint by adding $5cm^3$ ($\sim 0.005\ dm^3$) of the same $H_2SO_4(aq)$ in excess, determine the pH of the solution in the flask. (9)

1.2 $V(excess) = 0.005\ dm^3$

$V_{total} = V_{neutralisation} + V_{excess} = 0.0125 + 0.005 = 0.0175\ dm^3$ } H_2SO_4

$H_2SO_4: c = \frac{n}{V} \Rightarrow n = c \times V = 0.1 \times 0.0175 = 0.00175\ mol$ *

$c = \frac{n}{V_{excess}} \Rightarrow n = 0.1 \times 0.005 = 5 \times 10^{-4}\ mol$

$H_2SO_4 + 2H_2O \rightarrow 2H_3O^+ + SO_4^{2-}$

$H_2SO_4 : 2H_3O^+$

$1 : 2$

$5 \times 10^{-4} : 0.001\ (mol)$

$V_{...} = V(NaOH) + V(total, H_2SO_4) = 0.025 + 0.0175 = 0.0425\ dm^3$

$V_{total} = V(NaOH) + V(total, H_2SO_4) = 0.025 + 0.0175 = 0.0425\ dm^3$

$[H_3O^+] = \frac{n}{V_{total}} = \frac{0.001}{0.0425} = 0.2353\ mol.dm^{-3}$

$pH = -\log[H_3O^+] = -\log(0.2353) = 1.63$

ACIDS AND BASES

QUESTION 1: Multiple choice questions

1.1 Consider the equilibrium $H_2SO_4 + HSO_3^- \rightleftharpoons HSO_4^- + H_2SO_3$.
The two Lowry-Brønsted bases are: (*conjugates*)

A HSO_4^- and H_2SO_4
B HSO_3^- and HSO_4^-
C HSO_3^- and H_2SO_3
D HSO_4^- and H_2SO_3 (2)

1.2 Water can act as either an acid or a base. Which equation represents water reacting as an acid? (*donating proton*)

A $H_2O(\ell) + NH_3(g) \rightleftharpoons OH^-(aq) + NH_4^+(aq)$
B $H_2O(\ell) + HCl(aq) \rightleftharpoons H_3O^+(aq) + Cl^-(aq)$
C $H_2O(\ell) \rightleftharpoons H_2(g) + \frac{1}{2}O_2(g)$
D $H_2O(\ell) + C(s) \rightleftharpoons CO(g) + H_2(g)$ (2)

1.3 Consider the following equilibrium: $HC_2O_4^- + HCO_3^- \rightleftharpoons H_2CO_3 + C_2O_4^{2-}$
Which ONE of the following CORRECTLY identifies the order of Lowry-Brønsted acids and bases in the above reaction? (*conjugates*)

A Base, acid, acid, base
B Acid, base, base, acid
C Acid, base, acid, base
D Base, acid, base, acid (2)

1.4 Which ONE of the reactions below will produce the salt sodium ethanoate (sodium acetate)? (*acid and base salts formation*)

A $HCl(s) + CH_3COOH(aq) \rightarrow$
B $CH_3COOH(aq) + H_2O(\ell) \rightarrow$
C $CH_3COOH(aq) + NaOH(aq) \rightarrow$
D $H_2CO_3(aq) + NaOH(aq) \rightarrow$ (2)

1.5 If base P is titrated against acid Q, the pH of the solution at the end point is 9.
Which ONE of the following combinations CORRECTLY represents base P and acid Q?

	BASE P	ACID Q
A	KOH	HNO_3
B	$NaHCO_3$	HCl
C	NH_3	H_2SO_4
D	NaOH	CH_3COOH

(2)

2. NH_4^+ ions are mixed with HCO_3^- ions

2.1 Write a balanced equation for the reaction that takes place.

$NH_4^+ + HCO_3^- \rightleftharpoons NH_3 + H_2CO_3$

2.2 Define the term conjugate acid-base pair.

2.3 Identify the two bases in the above reaction.

HCO_3^- ...

2.3 Identify the two bases in the above reaction.

HCO_3^- and NH_3

2.4 The hydrogen sulphate ion can act as both an acid and a base. It reacts with water according to the following balanced equation.

$$HSO_4^- + H_2O \rightarrow H_2SO_4 + OH^-$$

2.4.1 Write down one word for the underlined sentence. Ampholyte

2.4.2 Show with the aid of equations that HSO_4^- can act as a base or acid.

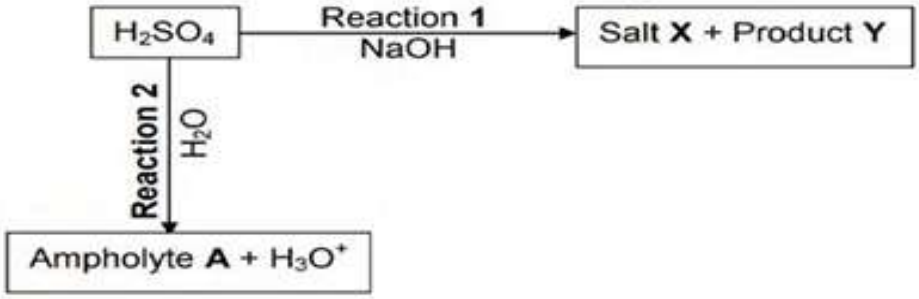
$HSO_4^-(aq) + H_2O(\ell) \rightleftharpoons SO_4^{2-}(aq) + H_3O^+(aq)$, acting as an acid

$HSO_4^-(aq) + H_2O(\ell) \rightleftharpoons H_2SO_4(aq) + OH^-(aq)$, acting as a base

[12]

QUESTION 3 (*interpretation and stoichiometry calculations*)

3.1 Two reactions of sulphuric acid are shown in the diagram below.



3.1.1 Define a Lowry-Brønsted base. (2)

3.1.2 Write down a balanced equation for Reaction 1 (3)

3.1.3 Write down the NAME of the salt represented by X (2)

3.1.4 Write down the FORMULA of ampholyte A (2)

3.1.5 Write down the formulae of the TWO conjugate acid-base pairs in Reaction 2. (4)

3.1.1 A Lowry-Brønsted base is proton acceptor.

3.1.2 $H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(\ell)$

3.1.3 Sodium sulphate

3.1.4 $H_2SO_4 + H_2O \rightarrow HSO_4^- + H_3O^+$, reaction

$\therefore HSO_4^-$

$\therefore HSO_4^-$

3.1.5 Pair 1: H_2SO_4 and HSO_4^-

Pair 2: H_3O^+ and H_2O